

Technical Document #278: Data Engineering - Comprehensive Overview

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Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Key Takeaways:

- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a data warehouse.
- Data quality ensures data is accurate, complete, and consistent.
- Data lakes store raw data in its native format.